👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 9 · STANDARD 6.AT.11

Explore Relationships Between Two Variables

Lesson 9-1 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can recognize when two quantities change together, and identify which one is the independent variable and which is the dependent variable.

Language Objective: I can explain which quantity depends on the other using the words independent variable, dependent variable, and depends on.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

independent variable · **dependent variable** · **depends on** · **quantity** · **relationship** · **Table of values** · **Input** · **Output**

Level 3: try the blanks from memory first, then check the bank.

1 The input quantity in a relationship is the _____ _____.

2. The quantity that changes in response to the other is the _____ _____.

3. Say it in order: the responding quantity _____ _____ the chosen one.

4. Something that can be counted or measured is a _____.

5. How two quantities change together is a _____.

6. A list of matched input and output pairs is a _____.

7. The independent value you put in is the _____.

8. The dependent value that comes out is the _____.

Write About the Math

Which two quantities in this shop are related, so that one depends on the other — and which quantity does NOT depend on the number of pizzas at all?
¿Qué dos cantidades de esta pizzería están relacionadas, de modo que una depende de la otra — y qué cantidad no depende para nada del número de pizzas?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **independent variable** *variable independiente*
- ☐ **dependent variable** *variable dependiente*
- ☐ **depends on** *depende de*
- ☐ **quantity** *cantidad*
- ☐ **relationship** *relación*

Model: *Akela chose where to place each tray, which set the hours of sunlight; the plant height was measured afterward, in response to the sunlight.* — A strong answer names sunlight as chosen (independent) and height as responding (dependent), and can say the sentence 'height depends on sunlight.' A weak answer just says 'height is bigger' without naming which quantity was chosen.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **In this scenario, the ___ is the independent variable. The ___ and the ___ both depend on it. The number of ___ stays the same no matter how many pizzas are made.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** independent variable
2. **Notes 2:** dependent variable
3. **Notes 3:** depends on
4. **Notes 4:** quantity
5. **Notes 5:** relationship
6. **Notes 6:** Table of values
7. **Notes 7:** Input

8. **Notes 8:** Output
9. **Watch for this mistake:** *Naming the variables backward — calling the responding quantity 'independent' because it is the one the question asks about.*
10. **Try It 1:** D. A student's height and their house number — *A house number is assigned by an address, not by how tall someone is — the two are unrelated.*
11. **Try It 2:** C. Pages printed depends on minutes running — *You choose how long to let it run; the number of pages is the result of that choice.*